

PRACTICAL WEEK -4

CREATE BOOKS TABLE

The screenshot shows the MySQL Workbench interface. The left sidebar contains the 'MANAGEMENT' section with options like Server Status, Client Connections, and Users and Privileges. The 'INSTANCE' section includes Startup / Shutdown, Server Logs, and Options File. The 'PERFORMANCE' section has Dashboard, Performance Reports, and Performance Schema Setup. The 'Administration' tab is selected, showing 'Schemas' and 'Information'. The 'Information' section displays 'No object selected'.

The main query editor shows the following SQL code:

```
1 • USE 2520080053_Ruhitha;
2 • CREATE TABLE Books
3
4 (
5     book_id INT AUTO_INCREMENT PRIMARY KEY,
6     title VARCHAR(200) NOT NULL,
7     author VARCHAR(150) NOT NULL,
8     genre VARCHAR(100),
9     description TEXT,
10    published_year INT
11 );
```

The 'Output' pane at the bottom shows the execution results:

#	Time	Action	Message
1	14:37:44	USE 2520080053_Ruhitha	0 row(s) affected
2	14:37:44	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT NULL, author VARCHAR(150) NOT NULL, genre VARCHAR(100), description TEXT, published_year INT)	0 row(s) affected

DESCRIBE Books;

The screenshot shows the MySQL Workbench interface after executing the DESCRIBE Books command. The left sidebar is the same as in the previous screenshot. The main query editor shows the following SQL code:

```
4     book_id INT AUTO_INCREMENT PRIMARY KEY,
5     title VARCHAR(200) NOT NULL,
6     author VARCHAR(150) NOT NULL,
7     genre VARCHAR(100),
8     description TEXT,
9     published_year INT
10 );
11 • DESCRIBE Books;
```

The 'Result Grid' pane shows the table structure:

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(200)	NO		NULL	
author	varchar(150)	NO		NULL	
genre	varchar(100)	YES		NULL	
description	text	YES		NULL	
published_year	int	YES		NULL	

The 'Output' pane at the bottom shows the execution results:

#	Time	Action	Message
1	14:37:44	USE 2520080053_Ruhitha	0 row(s) affected
2	14:37:44	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT NULL, author VARCHAR(150) NOT NULL, genre VARCHAR(100), description TEXT, published_year INT)	0 row(s) affected
3	14:39:17	DESCRIBE Books	6 row(s) returned

INSERT BOOK DATA

2520080053_Ruhitha

File Edit View Query Database Server Tools Scripting Help

Navigator: Query 1 SQL File 3* week3 joints and sets SQL File 5* x

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

```

82 'Orson Scott Card',
83 'Science Fiction',
84 'A young student is trained for an interstellar conflict.',
85 1985
86 ),
87
88 (
89 'Harry Potter and the Sorcerers Stone',
90 'J.K. Rowling',
91 'Fantasy',
92 'A young wizard begins his magical education.',
93 1997
94 ),
95
96 (
97 'Twenty Thousand Leagues Under the Seas',
98 'Jules Verne',
99 'Adventure',
100 'A fantastic underwater adventure aboard a mysterious submarine.',
101 1870
102 );

```

Output

Action Output

#	Time	Action	Message
1	14:37:44	USE 2520080053_Ruhitha	0 row(s) affected
2	14:37:44	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT NU...	0 row(s) affected
3	14:39:17	DESCRIBE Books	6 row(s) returned
4	14:46:01	INSERT INTO Books (title, author, genre, description, published_year) VALUES ('The Martian',...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0

DISPLAY BOOKS

2520080053_Ruhitha

File Edit View Query Database Server Tools Scripting Help

Navigator: Query 1 SQL File 3* week3 joints and sets SQL File 5* x

MANAGEMENT

- Server Status
- Client Connections
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- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

```

104 SELECT
105     book_id,
106     title,
107     author,
108     genre,
109     published_year
110 FROM Books
111 ORDER BY book_id;

```

Result Grid

	book_id	title	author	genre	published_year
1		The Martian	Andy Weir	Science Fiction	2011
2		Dune	Frank Herbert	Science Fiction	1965
3		The Hitchhikers Guide to the Galaxy	Douglas Adams	Science Fiction	1979
4		Pride and Prejudice	Jane Austen	Romance	1813
5		The Great Gatsby	F. Scott Fitzgerald	Classic	1925
6		Foundation	Isaac Asimov	Science Fiction	1951
7		The Hobbit	J.R.R. Tolkien	Fantasy	1937
8		Enders Game	Orson Scott Card	Science Fiction	1985
9		Harry Potter and the Sorcerers Stone	J.K. Rowling	Fantasy	1997
10		Twenty Thousand Leagues Under the Seas	Jules Verne	Adventure	1870

Books 2 x

Output

Action Output

#	Time	Action	Message
1	14:37:44	USE 2520080053_Ruhitha	0 row(s) affected
2	14:37:44	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT NU...	0 row(s) affected
3	14:39:17	DESCRIBE Books	6 row(s) returned
4	14:46:01	INSERT INTO Books (title, author, genre, description, published_year) VALUES ('The Martian',...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
5	15:18:47	SELECT book_id, title, author, genre, published_year FROM Books ORDER BY book_id LIMIT 0, 1...	10 row(s) returned

Object Info Session

3 30°C Partly sunny

Search

ADD EMBEDDING COLUMN

The screenshot shows the SQL Developer interface with a query window containing the following SQL code:

```
109 published_year
110 FROM Books
111 ORDER BY book_id;
112
113
114 • ALTER TABLE Books
115   ADD COLUMN embedding JSON;
116 • DESCRIBE Books;
```

Below the query window, the 'Result Grid' displays the table structure for 'Books':

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI		auto_increment
title	varchar(200)	NO			
author	varchar(150)	NO			
genre	varchar(100)	YES			
description	text	YES			
published_year	int	YES			
embedding	json	YES			

The 'Output' window shows the execution results:

#	Time	Action	Message
1	14:55:58	ALTER TABLE Books ADD COLUMN embedding JSON	Error Code: 1046. No database selected Select the default DB to be used by double-clicking its name in the
2	14:56:59	USE 2520080053_Ruhitha	0 row(s) affected
3	14:57:05	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
4	14:57:05	DESCRIBE Books	7 row(s) returned

PART 7 - INSERT MOCK VECTORS

BOOK 1 - THE MARTIAN

The screenshot shows the SQL Developer interface with a query window containing the following SQL code:

```
100 'A fantastic underwater adventure aboard a mysterious submarine.',
101 1870
102 );
103
104 • SELECT
105   book_id,
106   title,
107   author,
108   genre,
109   published_year
110 FROM Books
111 ORDER BY book_id;
112
113
114 • ALTER TABLE Books
115   ADD COLUMN embedding JSON;
116 • DESCRIBE Books;
117
118 • UPDATE Books
119   SET embedding = JSON_ARRAY(0.90, 0.80, 0.20)
120   WHERE book_id = 1;
```

The 'Output' window shows the execution results:

#	Time	Action	Message
1	14:55:58	ALTER TABLE Books ADD COLUMN embedding JSON	Error Code: 1046. No database selected
2	14:56:59	USE 2520080053_Ruhitha	0 row(s) affected
3	14:57:05	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
4	14:57:05	DESCRIBE Books	7 row(s) returned
5	15:04:12	UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1	1 row(s) affected Rows matched: 1

BOOK 2 - DUNE

File Edit View Query Database Server Tools Scripting Help

Navigator: Query 1 SQL File 3* week3 joints and sets SQL File 5*

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

```

104 SELECT
105     book_id,
106     title,
107     author,
108     genre,
109     published_year
110 FROM Books
111 ORDER BY book_id;
112
113
114 ALTER TABLE Books
115 ADD COLUMN embedding JSON;
116 DESCRIBE Books;
117
118 UPDATE Books
119 SET embedding = JSON_ARRAY(0.90, 0.80, 0.20)
120 WHERE book_id = 1;
121
122 UPDATE Books
123 SET embedding = JSON_ARRAY(0.85, 0.75, 0.25)
124 WHERE book_id = 2;

```

Output

Action Output

#	Time	Action	Message
2	14:56:59	USE 2520080053_Ruhitha	0 row(s) affected
3	14:57:05	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
4	14:57:05	DESCRIBE Books	7 row(s) returned
5	15:04:12	UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
6	15:05:06	UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

Object Info Session

BOOK 3 - HITCHHIKERS GUIDE

File Edit View Query Database Server Tools Scripting Help

Navigator: 2520080053_Ruhitha x

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

```

108     genre,
109     published_year
110 FROM Books
111 ORDER BY book_id;
112
113
114 ALTER TABLE Books
115 ADD COLUMN embedding JSON;
116 DESCRIBE Books;
117
118 UPDATE Books
119 SET embedding = JSON_ARRAY(0.90, 0.80, 0.20)
120 WHERE book_id = 1;
121
122 UPDATE Books
123 SET embedding = JSON_ARRAY(0.85, 0.75, 0.25)
124 WHERE book_id = 2;
125
126 UPDATE Books
127 SET embedding = JSON_ARRAY(0.80, 0.70, 0.30)
128 WHERE book_id = 3;

```

Output

Action Output

#	Time	Action	Message
3	14:57:05	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
4	14:57:05	DESCRIBE Books	7 row(s) returned
5	15:04:12	UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
6	15:05:06	UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
7	15:05:52	UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

Object Info Session

BOOK 4 - PRIDE AND PREJUDICE

SQL File 5*

```

113
114 • ALTER TABLE Books
115 • ADD COLUMN embedding JSON;
116 • DESCRIBE Books;
117
118 • UPDATE Books
119   SET embedding = JSON_ARRAY(0.90, 0.80, 0.20)
120   WHERE book_id = 1;
121
122 • UPDATE Books
123   SET embedding = JSON_ARRAY(0.85, 0.75, 0.25)
124   WHERE book_id = 2;
125
126 • UPDATE Books
127   SET embedding = JSON_ARRAY(0.80, 0.70, 0.30)
128   WHERE book_id = 3;
129
130 • UPDATE Books
131   SET embedding = JSON_ARRAY(0.10, 0.20, 0.90)
132   WHERE book_id = 4;
133

```

Output

#	Time	Action	Message
4	14:57:05	DESCRIBE Books	7 row(s) returned
5	15:04:12	UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
6	15:05:06	UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
7	15:05:52	UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
8	15:06:35	UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

BOOK 5 – GREAT GATSBY

SQL File 5*

```

116 • DESCRIBE Books;
117
118 • UPDATE Books
119   SET embedding = JSON_ARRAY(0.90, 0.80, 0.20)
120   WHERE book_id = 1;
121
122 • UPDATE Books
123   SET embedding = JSON_ARRAY(0.85, 0.75, 0.25)
124   WHERE book_id = 2;
125
126 • UPDATE Books
127   SET embedding = JSON_ARRAY(0.80, 0.70, 0.30)
128   WHERE book_id = 3;
129
130 • UPDATE Books
131   SET embedding = JSON_ARRAY(0.10, 0.20, 0.90)
132   WHERE book_id = 4;
133 • UPDATE Books
134   SET embedding = JSON_ARRAY(0.15, 0.30, 0.80)
135   WHERE book_id = 5;
136

```

Output

#	Time	Action	Message
5	15:04:12	UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1	1 row(s) affect
6	15:05:06	UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2	1 row(s) affect
7	15:05:52	UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3	1 row(s) affect
8	15:06:35	UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4	1 row(s) affect
9	15:10:17	UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5	1 row(s) affect

BOOK 6 – FOUNDATION

2520080053_Ruhiha x

File Edit View Query Database Server Tools Scripting Help

Navigator: Query 1 SQL File 3* week3 joints and sets SQL File 5* x

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Object Info Session

```

119 SET embedding = JSON_ARRAY(0.90, 0.80, 0.20)
120 WHERE book_id = 1;
121
122 • UPDATE Books
123 SET embedding = JSON_ARRAY(0.85, 0.75, 0.25)
124 WHERE book_id = 2;
125
126 • UPDATE Books
127 SET embedding = JSON_ARRAY(0.80, 0.70, 0.30)
128 WHERE book_id = 3;
129
130 • UPDATE Books
131 SET embedding = JSON_ARRAY(0.10, 0.20, 0.90)
132 WHERE book_id = 4;
133 • UPDATE Books
134 SET embedding = JSON_ARRAY(0.15, 0.30, 0.80)
135 WHERE book_id = 5;
136
137 • UPDATE Books
138 SET embedding = JSON_ARRAY(0.95, 0.65, 0.20)
139 WHERE book_id = 6;

```

Output

#	Time	Action	Message
6	15:05:06	UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
7	15:05:52	UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
8	15:06:35	UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
9	15:10:17	UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
10	15:11:44	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

BOOK 7 - THE HOBBIT

File Edit View Query Database Server Tools Scripting Help

Navigator: Query 1 SQL File 3* week3 joints and sets SQL File 5* x

MANAGEMENT

- Server Status
- Client Connections
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- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

```

123 SET embedding = JSON_ARRAY(0.85, 0.75, 0.25)
124 WHERE book_id = 2;
125
126 • UPDATE Books
127 SET embedding = JSON_ARRAY(0.80, 0.70, 0.30)
128 WHERE book_id = 3;
129
130 • UPDATE Books
131 SET embedding = JSON_ARRAY(0.10, 0.20, 0.90)
132 WHERE book_id = 4;
133 • UPDATE Books
134 SET embedding = JSON_ARRAY(0.15, 0.30, 0.80)
135 WHERE book_id = 5;
136
137 • UPDATE Books
138 SET embedding = JSON_ARRAY(0.95, 0.65, 0.20)
139 WHERE book_id = 6;
140
141 • UPDATE Books
142 SET embedding = JSON_ARRAY(0.30, 0.75, 0.90)
143 WHERE book_id = 7;

```

Output

#	Time	Action	Message
1	15:13:55	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

BOOK 8 – ENDERS GAME

The screenshot shows the DBeaver SQL editor interface. The left sidebar contains navigation tabs for MANAGEMENT, INSTANCE, PERFORMANCE, Administration, and Information. The main editor area displays a SQL query with line numbers 128 to 148. The query consists of several UPDATE statements for a 'Books' table, each setting a 'embedding' field to a JSON array and filtering by 'book_id'. The output pane at the bottom shows two successful execution results.

```
128 WHERE book_id = 3;
129
130 • UPDATE Books
131 SET embedding = JSON_ARRAY(0.10, 0.20, 0.90)
132 WHERE book_id = 4;
133 • UPDATE Books
134 SET embedding = JSON_ARRAY(0.15, 0.30, 0.80)
135 WHERE book_id = 5;
136
137 • UPDATE Books
138 SET embedding = JSON_ARRAY(0.95, 0.65, 0.20)
139 WHERE book_id = 6;
140
141 • UPDATE Books
142 SET embedding = JSON_ARRAY(0.30, 0.75, 0.90)
143 WHERE book_id = 7;
144
145 • UPDATE Books
146 SET embedding = JSON_ARRAY(0.88, 0.85, 0.15)
147 WHERE book_id = 8;
148
```

Output

#	Time	Action	Message
✓ 1	15:13:55	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 2	15:14:32	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

BOOK 9 – HARRY POTTER

The screenshot shows the DBeaver SQL editor interface. The left sidebar contains navigation tabs for MANAGEMENT, INSTANCE, PERFORMANCE, Administration, and Information. The main editor area displays a SQL query with line numbers 131 to 151. The query consists of several UPDATE statements for a 'Books' table, each setting a 'embedding' field to a JSON array and filtering by 'book_id'. The output pane at the bottom shows three successful execution results.

```
131 SET embedding = JSON_ARRAY(0.10, 0.20, 0.90)
132 WHERE book_id = 4;
133 • UPDATE Books
134 SET embedding = JSON_ARRAY(0.15, 0.30, 0.80)
135 WHERE book_id = 5;
136
137 • UPDATE Books
138 SET embedding = JSON_ARRAY(0.95, 0.65, 0.20)
139 WHERE book_id = 6;
140
141 • UPDATE Books
142 SET embedding = JSON_ARRAY(0.30, 0.75, 0.90)
143 WHERE book_id = 7;
144
145 • UPDATE Books
146 SET embedding = JSON_ARRAY(0.88, 0.85, 0.15)
147 WHERE book_id = 8;
148
149 • UPDATE Books
150 SET embedding = JSON_ARRAY(0.20, 0.55, 0.95)
151 WHERE book_id = 9;
```

Output

#	Time	Action	Message
✓ 1	15:13:55	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 2	15:14:32	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 3	15:15:11	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

BOOK 10 – TWENTY THOUSAND LEAGUES

File Edit View Query Database Server Tools Scripting Help

Navigator: Query 1 SQL File 3* week3 joints and sets SQL File 5* x

Limit to 1000 rows

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

```

133 • UPDATE Books
134   SET embedding = JSON_ARRAY(0.15, 0.30, 0.80)
135   WHERE book_id = 5;
136
137 • UPDATE Books
138   SET embedding = JSON_ARRAY(0.95, 0.65, 0.20)
139   WHERE book_id = 6;
140
141 • UPDATE Books
142   SET embedding = JSON_ARRAY(0.30, 0.75, 0.90)
143   WHERE book_id = 7;
144
145 • UPDATE Books
146   SET embedding = JSON_ARRAY(0.88, 0.85, 0.15)
147   WHERE book_id = 8;
148
149
150
151 • UPDATE Books
152   SET embedding = JSON_ARRAY(0.70, 0.90, 0.40)
153   WHERE book_id = 10;
  
```

Output

Action Output

#	Time	Action	Message
1	15:13:55	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
2	15:14:32	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
3	15:15:11	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
4	15:16:12	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

PART 8 - VERIFY ALL VECTORS

File Edit View Query Database Server Tools Scripting Help

Navigator: Query 1 SQL File 3* week3 joints and sets SQL File 5* x

Limit to 1000 rows

MANAGEMENT

- Server Status
- Client Connections
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- Data Import/Restore

INSTANCE

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- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

```

148
149
150
151 • UPDATE Books
152   SET embedding = JSON_ARRAY(0.70, 0.90, 0.40)
153   WHERE book_id = 10;
154
155 • SELECT
156   book_id,
  
```

Result Grid

book_id	title	embedding
1	The Martian	[0.90, 0.80, 0.20]
2	Dune	[0.85, 0.75, 0.25]
3	The Hitchhikers Guide to the Galaxy	[0.80, 0.70, 0.30]
4	Pride and Prejudice	[0.10, 0.20, 0.90]
5	The Great Gatsby	[0.15, 0.30, 0.80]
6	Foundation	[0.95, 0.65, 0.20]
7	The Hobbit	[0.30, 0.75, 0.90]
8	Enders Game	[0.88, 0.85, 0.15]
9	Harry Potter and the Sorcerers Stone	[0.20, 0.55, 0.95]
10	Twenty Thousand Leagues Under the Seas	[0.70, 0.90, 0.40]

Books 2 x

Output

Action Output

#	Time	Action	Message
1	15:13:55	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
2	15:14:32	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
3	15:15:11	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
4	15:16:12	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
5	15:16:55	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1000	10 row(s) returned

PART 9 - SEARCH VECTOR

The instructor provides a search vector.

For this practical we use:

[0.90, 0.80, 0.20]

This represents:

Stories about space

PART 10 - L2 DISTANCE

The screenshot shows a database management interface with a query editor and a results grid. The query editor displays the following SQL code:

```
193      ),
194      4
195    ) AS l2_distance
196 FROM Books
197 WHERE embedding IS NOT NULL
198 ORDER BY l2_distance ASC
199 LIMIT 3;
```

The results grid shows the following data:

book_id	title	author	genre	l2_distance
1	The Martian	Andy Weir	Science Fiction	0
8	Enders Game	Orson Scott Card	Science Fiction	0.0735
2	Dune	Frank Herbert	Science Fiction	0.0866

The interface also shows a sidebar with navigation options and a bottom section with an "Action Output" table.

#	Time	Action	Message
2	15:14:32	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
3	15:15:11	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
4	15:16:12	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
5	15:16:55	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1000	10 row(s) returned
6	15:19:21	SELECT book_id, title, author, genre, ROUND(SQRT(POW(CAST(...	3 row(s) returned

PART 11 - COSINE SIMILARITY

FileEditViewQueryDatabaseServerToolsScriptingHelp

Navigator

MANAGEMENT

Server Status

Client Connections

Users and Privileges

Status and System Variables

Data Export

Data Import/Restore

INSTANCE

Startup / Shutdown

Server Logs

Options File

PERFORMANCE

Dashboard

Performance Reports

Performance Schema Setup

AdministrationSchemas

No object selected

Query 1SQL File 3*week3 joints and setsSQL File 5*×

Limit to 1000 rows

259

260

261

262

263

264

265

266

Result Grid

Filter Rows:

Export:

Wrap Cell Contents:

Fetch rows:

	book_id	title	author	genre	cosine_similarity
1	1	The Martian	Andy Weir	Science Fiction	1
2	2	Dune	Frank Herbert	Science Fiction	0.9986
8	8	Enders Game	Orson Scott Card	Science Fiction	0.9983

Result 4 ×

Output

Action Output

#	Time	Action	Message
3	15:15:11	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
4	15:16:12	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
5	15:16:55	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1000	10 row(s) returned
6	15:19:21	SELECT book_id, title, author, genre, ROUND(SQRT(POW(CAST(...	3 row(s) returned
7	15:26:41	SELECT book_id, title, author, genre, ROUND((CAST(JSON_EXTRACT(...	3 row(s) returned